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erDiagram
    USER_ACCOUNT {
        int user_id PK
        varchar username UK
        varchar password
        enum role
        int related_id
        enum status
        datetime created_time
    }

    STUDENT {
        int student_id PK
        varchar student_no UK
        varchar name
        enum gender
        varchar college
        varchar major
        varchar class
        varchar phone
        varchar email
        enum check_status
    }

    APARTMENT_MANAGER {
        int manager_id PK
        varchar employee_no UK
        varchar name
        enum gender
        varchar phone
        varchar email
        enum status
    }

    APARTMENT {
        int apartment_id PK
        varchar apartment_name UK
        varchar address
        int floor_count
        int manager_id FK
        text description
    }

    ROOM {
        int room_id PK
        int apartment_id FK
        varchar room_no
        int floor
        enum room_type
        int capacity
        int current_count
    }
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        enum status
    }

ACCOMMODATION {
    int accommodation_id PK
    int student_id FK
    int room_id FK
    int bed_no
    date check_in_date
    date check_out_date
    enum status
}

REPAIR_REQUEST {
    int repair_id PK
    int student_id FK
    int room_id FK
    varchar repair_type
    text description
    varchar image_path
    enum status
    datetime submit_time
    datetime handle_time
    int handler_id FK
    text result
    varchar result_image_path
}

VISITOR_LOG {
    int visitor_id PK
    varchar visitor_name
    varchar visitor_phone
    varchar id_card
    int student_id FK
    int room_id FK
    varchar visit_reason
    datetime visit_time
    datetime leave_time
    int register_manager_id FK
    text remark
}

NOTICE {
    int notice_id PK
    varchar title
    text content
    varchar image_path
    varchar file_path
    varchar video_path
    datetime publish_time
    int publisher_id FK
}

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SYSTEM_LOG {
    int log_id PK
    int user_id FK
    varchar operation
    datetime operation_time
    varchar ip_address
    text description
}

APARTMENT_MANAGER ||--o{ APARTMENT : manages
APARTMENT ||--o{ ROOM : contains

STUDENT ||--o{ ACCOMMODATION : has
ROOM ||--o{ ACCOMMODATION : provides

STUDENT ||--o{ REPAIR_REQUEST : submits
ROOM ||--o{ REPAIR_REQUEST : belongs_to
APARTMENT_MANAGER ||--o{ REPAIR_REQUEST : handles

STUDENT ||--o{ VISITOR_LOG : visited_by
ROOM ||--o{ VISITOR_LOG : visit_room
APARTMENT_MANAGER ||--o{ VISITOR_LOG : registers

APARTMENT_MANAGER ||--o{ NOTICE : publishes

USER_ACCOUNT ||--o{ SYSTEM_LOG : generates
```